CLASS 8 — SCIENCE CHAPTER TEST SET A

Chapter 9: **The Amazing World of Solutes, Solvents and Solutions** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A substance has mass 200 g and volume 50 cm³. Its density is: 1

(a) 8 g/cm³ **(b)** 0.25 g/cm³ **(c)** 2 g/cm³ **(d)** 4 g/cm³

2. A solution is a: 1

(a) non-uniform mixture **(b)** compound
(c) pure element **(d)** uniform mixture

3. A solution in which more solute can still dissolve is: 1

(a) insoluble **(b)** supersaturated only
(c) unsaturated **(d)** saturated

4. Solubility of gases in water with rise in temperature: 1

(a) increases **(b)** decreases **(c)** doubles **(d)** stays same

5. Volume is the: 1

(a) force of Earth **(b)** matter in object
(c) heat of object **(d)** space occupied by an object

6. Density is defined as: 1

(a) force per area **(b)** mass per unit volume
(c) weight per unit area **(d)** volume per unit mass

7. An object sinks in water if its density is: 1

(a) equal to ice **(b)** more than water **(c)** zero **(d)** less than water

8. Generally, density with a rise in temperature: 1

(a) increases **(b)** stays same **(c)** doubles **(d)** decreases

9. A solution of sugar in water is classified as: 1

(a) a compound **(b)** a pure substance
(c) an element **(d)** a homogeneous mixture

10. The solubility of sugar at 30 °C is 220 g per 100 mL. If 250 g is added to 100 mL, undissolved sugar is: 1

(a) 220 g **(b)** 0 g **(c)** 30 g **(d)** 250 g

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Define a solution with an example. 2

12. Differentiate solute from solvent. 2

13. Define saturated and unsaturated solutions. 2

14. Define solubility. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Explain with an activity how to know whether a solution is saturated. 3

16. Describe how temperature changes the solubility of sugar in water. 3

17. A student dissolves 36 g of salt in 100 mL of water at 20 °C and finds saturation. What is the solubility of the salt? What happens if 10 g more is added? 3

OR

Distinguish between saturated, unsaturated and dilute solutions. 3

18. Explain why the density of a substance decreases when it is heated. 3

OR

The mass of a stone is 120 g and its volume is 40 cm^3. Calculate density. Will it float in water? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. Explain solute, solvent, solution, saturated, unsaturated and solubility with examples. 5

OR

Describe an activity to determine the solubility of salt in water at room temperature. 5

20. Explain the effect of temperature on solubility of solids and gases with two examples each. 5

OR

Define density. Describe an experiment to find the density of a stone or a small block. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND